

Hw3

10. $C(x)$ = student x has a cat

$D(x)$ = student x has a dog

$F(x)$ = student x has a ferret

Answers:

a) $\exists x (C(x) \wedge D(x) \wedge F(x))$

b) $\forall x (C(x) \wedge D(x) \wedge F(x))$

c) $\exists x C(x) \wedge \exists x D(x) \wedge \exists x F(x)$ or $\exists x C(x) \exists y D(y) \exists z F(z)$

15. Let $S(x, y)$ be x is shorter than y .

Given $\exists s S(s, \text{Lenier})$, it follows that $S(\text{Lenier}, \text{Lenier})$.

Then by existential generalization it follows that

$\exists s S(s, s)$, so that someone is shorter than himself.

Answer:

The error is in the step from $\exists s S(s, \text{Lenier})$ to $S(\text{Lenier}, \text{Lenier})$.

In this step, the existential quantifier asserts that there exists some individual (like i) such that $S(i, \text{Lenier})$. It doesn't

imply that Lenier himself is shorter than Lenier. For this

statement specifically, this would also be implying Lenier

is shorter than himself, which is false. So it's not

possible to instantiate or generalize the existential quantifier about

height to Lenier unless it's known that Lenier is the one

shorter than Lenier (which he isn't). The correct instantiation

would be to another individual. $\rightarrow S(i, \text{Lenier})$. The argument is

invalid because the existential only guarantees some person is shorter

than Lenier, not that Lenier is shorter than himself.